

To Boldly Go Where No One Has Gone Before¹

Your name:

Who you are working with:



Today we are going to review what we did last week in order to try to grasp exactly how unfathomably big space is. This will help us tomorrow when we look at the Drake² Equation which allows us estimate how many intelligent alien species there are in our galaxy (and the observable universe overall).

Warm Up—10 Minutes

For the following questions you will need the fact that Light Travels at a whopping 299 792 458 meters per second in a vacuum.³ Have fun!

1. How would you write the speed of light in scientific notation?

Sample Answer: $2,99792458 \times 10^8 m/s$

2. If you round to the second decimal place and change our units for kilometers per second what would our rounded speed of light be? (This number might be a bit easier to work with, no?)

Sample Answer: $3,00 \times 10^8 m/s \rightarrow 3,00 \times 10^5 km/s$

AB1 elementare Lösungsverfahren, direktes Anwenden von Formeln etc.

3. Earth has a circumference of approximately $40075 km$, approximately how many times can light circle the earth in one second?⁴

Sample Answer: $40075 km = 4.0075 \times 10^5$ from our previous answer, we have the speed of light divided by the diameter of the earth to get the number of times light could go around the earth per second (let's just round to the nearest hundreth here).

$$\frac{3,00 \times 10^6}{4,01 \times 10^5} = \frac{0,748 \times 10^1}{1} = 7,48(times/s)$$

approximately 7 and a half times!

AB1 elementare Lösungsverfahren, direktes Anwenden von Formeln etc.

¹Part 4 of an introduction to Scientific Notation and Scale in five parts, written by Joshua Aaron, Comic by Bill Waterson.

²No, unfortunately not that drake!

³Let's treat space like a vacuum for now... it mostly is, except for the billions upon billions of stars and planets and stuff...

⁴Getting light to circle around the earth like this is actually pretty difficult, though we get close when we send messages over the internet around the world through fiber optic cable!

4. How many meters does light go in one minute? What is that number expressed in kilometers? (We call this a *light-minute*, which, although it has minute in the name, measures distance—the distance time goes in a minute, does that make sense?.)

Sample Answer: $60s \cdot 3,00 \times 10^5 km/s = 1,80 \times 10^7 km$

AB1 elementare Lösungsverfahren, direktes Anwenden von Formeln etc.

Review

Last week we covered three super powerful and useful ideas when it came to exponents:

1. Multiplication and subtraction of of two exponents with the same base:

$$10^a \cdot 10^b = 10^{a+b} \text{ and } \frac{10^a}{10^b} = 10^{a-b}$$

2. Exponentiation of exponents: $(10^a)^b = 10^{a \cdot b} = 10^{ab}$

3. Negative Exponents: $10^{-a} = 10^{-1 \cdot a} = \frac{1}{10^a}$

On Thursday we began thinking about scientific notation, where we began to represent our numbers with digits $dcb a, cba$ as $d, cbacba \times 10^3$. We also began multiplying and dividing numbers in scientific notation, where we try to maintain our “one digit before the decimal point” rule whenever we can: so that $6,54 \times 10^4 \cdot 9,87 \times 10^8 = 64,55 \times 10^{12} = 6,455 \times 10^{13}$. With those in mind, let’s get to some mind blowing questions!

Practice: One Small Step For Student

1. Given that the Earth is approximately 149 600 000 kilometers (or $1,496 \times 10^8 km$) from the Sun, if the Sun were to spontaneously disappear⁵ approximately how long would it take for us here on Earth to notice? This distance (measured in light minutes, but also kilometers) is called 1 Astronomical Unit (or AU)

Sample Answer: Distance from the sun is $1,496 \times 10^8$ divided by 1 light minute: $1,80 \times 10^7 km/m$ gives us

$$\frac{1,496 \times 10^8}{1,80 \times 10^7} = 0,831 \times 10^1 = 8.31 min$$

AB1 elementare Lösungsverfahren, direktes Anwenden von Formeln etc.

2. Voyager 1, the farthest human-made object from Earth, is currently around 117 AU: How far is that in Kilometers? Light minutes?

Sample Answer: $117 \times 1,496 \cdot 10^8 km = 1,750 \times 10^{10} km$

For light minutes we divide by our previous answer:

$$\frac{1,750 \times 10^{10} km}{1,80 \times 10^7 km/min} = 0,972 \times 10^3 = 9,72 \times 10^2 min$$

Just for fun, this is around 972 light minutes or 16,2 light hours! (a signal from voyager 1 takes 16+ hours to reach us!)

AB1 elementare Lösungsverfahren, direktes Anwenden von Formeln etc.

⁵Don’t worry, it is highly unlikely that this will happen, that is, so far as we know...

3. The 3 billion base pairs that form the double helix of DNA in each of our cells is so tightly curled up you can't see them with the naked eye⁶ But if you were to stretch the helix out—one cell's worth of DNA is 2 meters long! Given that we have 37,2 *trillion* cells, if we were to stretch out all the DNA in all of our cells, how many kilometers would the resulting (messy) chain be? How does this relate to what you calculated in the previous question? Please express your answer in scientific notation!

Sample Answer: First let's translate 37,2 trillion into $37,2 \times 10^{12}$ then we multiply by two meters and divide by our AU from the previous question:

$$\frac{7,24 \times 10^{13}m}{1,750 \times 10^{10}km} = \frac{7,24 \times 10^{13}m}{1,750 \times 10^{13}m} = 4,14$$

This means if we were to uncurl the DNA in all of our cells it would stretch to more than four times the distance the farthest man made object has traveled! WHOA!

4. Above and Beyond Questions:

- (a) A slightly more mundane, but still very real and astronomical fact: how many seconds are there in a year? (What would your answer look like in scientific notation? This will be useful for the next question!)

Sample Answer: $60s \cdot 60min \cdot 24h \times 365 = 31536000 = 3,1536 \times 10^7 s/annum$ **AB1: Kennen+Anwenden von Definitionen, Regeln, Formeln, Algorithmen—leichte erfolgserlebnis**

- (b) When dealing with interstellar distances we usually use the distance measure of a *light-year*, measured by how far light can travel in a year. Given what we know, how would you calculate how many meters (or kilometers) are in a light year?⁷

Sample Answer: $3,1536 \times 10^7 s/annum \cdot 3.00 \times 10^8 m/s = 9,4608 \times 10^{15}m = 9,4608 \times 10^{12}km$ **AB1: Kennen+Anwenden von Definitionen, Regeln, Formeln, Algorithmen**

- (c) the nearest star to our solar system is Proxima Centauri at 4.25 light years away: What fraction of that distance has Voyager 1 Traveled? (*Hint:* Use your answer to question 2!) Please express your answer in scientific notation!

Sample Answer: $\frac{1,750 \times 10^{10}km \text{ Voyager distance}}{4,25 \cdot 9,4608 \times 10^{12} \text{ Proxima Centauri distance}} = 0,04352 \times 10^{-2} = 4,352 \times 10^{-4}$ as a percent this is 0,04352% (which is Nuthin!) About 4 Ten Thousandths! **AB2 mehrschrittiges Vorgehen**

⁶(they are only 6 microns across, $6 \times 10^{-6}m$!)

⁷Our Galaxy, the Milky Way, is between 100 000 and 180 000 light-years in diameter (as it is a spiral galaxy it doesn't have a consistent radius like a circle). We will come back to this tomorrow!